Predicting Vehicle Maintenance Needs Using Machine Learning

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Abstract—Vehicle maintenance is important for making vehicles more reliable and safer to use. It also helps in making the vehicle work better and more efficiently. This project looks at using machine learning to predict if a vehicle needs maintenance. The prediction is based on information about the vehicle and past maintenance records. A dataset with 50,000 records was used to create a model that can classify if maintenance is needed. The data was split into parts for training, checking and testing. The data was. Prepared by filling in missing numbers and scaling them. Also missing categories were. Converted into numbers using a method called one-hot encoding. A Logistic Regression model was. Tested. This model was given weights to handle the classes better. On a test set with 10,000 records the model had 95.95 percent accuracy and 97.34 percent balanced accuracy. The results show that the model works with the data it was given.. There is a strong link between the problems reported and what the model is trying to predict. This link needs study. The results should be seen as specific, to the dataset used. They need checking before they can be used in other situations.

Keywords: Predictive maintenance; Machine learning; Logistic regression; Vehicle maintenance; Classification; Confusion matrix; Feature engineering; Model evaluation

1. Introduction

Vehicle maintenance is very important for making sure vehicles are safe to use and work properly. It also helps control the cost of running a vehicle. Traditional ways of doing maintenance might use set times for service checks done by hand or the knowledge of people who own or fix the vehicles. These methods can sometimes lead to doing work than needed or not catching problems early enough.

Machine learning offers a way to look at details about a vehicle and past information to find patterns that show when maintenance is needed. Classification models can use the information to guess if a vehicle needs maintenance.

This project creates a machine learning model to predict when a vehicle needs maintenance using a vehicle maintenance dataset. The work involves cleaning the data training the model checking it with validation data testing it with a test set and looking at the results with a confusion matrix. The project also looks at whether the data has a balance, between types of cases and if the features used are good enough to make accurate predictions.

Problem Statement And Objectives

Problem Statement

Determining whether a vehicle requires maintenance using personal judgment can be difficult. This project investigates whether machine learning can separate vehicles into two categories: vehicles that need maintenance and vehicles that do not. The data used comes from a well-structured dataset.

Objectives

1. To prepare the vehicle maintenance dataset.

2. To build a classification model using Logistic Regression.

3. To evaluate the vehicle maintenance model performance with accuracy, accuracy, precision, recall, F1-score and a confusion matrix.

4. To examine how reported problems influence the vehicle maintenance model performance.

5. To identify issues that may complicate interpretation and practical use of vehicle maintenance results.

Scope

This is a prototype based on data, not a vehicle diagnostic system. The predictions come from patterns, in the labels and features provided. They are not proof of mechanical problems. They should not take the place of a technician doing a proper inspection..

2. Review of Literature

I have seen predictive maintenance grow over time into a method that uses machine learning in factories and cars. Machine learning looks at data to find patterns. It helps figure out when maintenance should be done.

In 2001 Breiman introduced Random Forests. This method combines decision trees to help with classification and prediction. Later in 2011 Pedregosa and colleagues described Scikit-learn. This is a library that provides tools for cleaning data, training models and checking how well the models perform. In 2021 Theissler and colleagues looked at how machine learning's used in maintenance. They explained how data-driven approaches help maintenance teams make decisions.

Studies, in this area show that good input features, reliable labels dealing with class imbalance and strong evaluation methods are all very important. Using this knowledge this project builds a workflow that classifies when cars need maintenance. It also tests how well the model works on a test set.

3. Research Methodology

3.1 Dataset Description

The project uses the Vehicle Maintenance Data dataset that is available on Kaggle. This dataset includes 50,000 records. The main thing we are trying to predict is called Need_Maintenance. This variable has two values. The value 0 means no maintenance is needed. The value 1 means maintenance is needed.

In the dataset there are 9,502 records where the value is 0. There are 40,498 records where the value's 1. This difference, in numbers shows that the data is not balanced. Because of this imbalance it is important to check how well the model works for both classes of just looking at accuracy.

3.2 Data Splitting

The dataset was split into training, validation and test groups using splitting and a fixed random state of 42 was used.The training set had thirty thousand records.The validation set had ten thousand records. The independent test set also had ten thousand records.

3.3 Data Preprocessing

Two raw date columns, Last_Service_Date and Warranty_Expiry_Date were left out from the input features. The other 22 input features had numerical and categorical information, such, as vehicle model, mileage, maintenance history reported issues, vehicle age, fuel type, transmission type, service history, accident history, vehicle-condition attributes and service-related date features.

Numerical features were handled using imputation and standardisation. Categorical features were handled using frequent-value imputation and one-hot encoding. The fitted preprocessing pipeline changed the input data into 40 model features. Preprocessing was trained on the training data. Then used on validation and test data.

3.4 Model Development

A logistic regression classifier was trained using the data that had been cleaned and prepared. The model had class_weight set to 'balanced' to handle the fact that the two groups in the target were not equally represented. The model was allowed to run for max_iter=2000 to make sure there were steps for the model to improve as much, as possible.

3.5 Model Evaluation

I evaluated the model with accuracy, accuracy, precision, recall, F1-score and a confusion matrix. I used validation data while I was developing the model. I kept the test set for the evaluation that I reported.

The confusion matrix shows how the actual labels match the predicted labels. The diagonal cells of the confusion matrix show the predictions while the cells, off the diagonal show the predictions.

4. Results / Findings

The final Logistic Regression model achieved 95.95% accuracy and 97.34% balanced accuracy on the test set of 10,000 records. The confusion matrix showed 1,892 negatives, 8 false positives, 397 false negatives and 7,703 true positives.

These results show that the model correctly classified records in both classes. It correctly identified 7,703 out of 8,100 records that were labeled as requiring maintenance. However 397 of those records were incorrectly labeled as not needing maintenance. This is a point because missing maintenance needs can lead to real-world problems.

The number of positives is very low meaning only 8 records that were labeled as not needing maintenance were wrongly predicted to require it. This suggests the model is conservative, in predicting maintenance needs. Still these results are based on the specific dataset and evaluation method used. More testing is needed before we can say how well the model will perform on vehicles not included in this data.

Table 1. Model Performance on the Test Dataset

Evaluation Metric	**Result**
Accuracy	95.95%
Balanced Accuracy	97.34%
Precision - Class 0	82.66%
Recall - Class 0	99.58%
F1-score - Class 0	90%
Precision - Class 1	99.90%
Recall - Class 1	95.10%
F1-score - Class 1	97%

Note: Class 0 represents vehicles not requiring maintenance, while Class 1 represents vehicles requiring maintenance.

Table 2. Confusion Matrix of the Proposed Model

Actual Class	Predicted: No Maintenance (0)	Predicted: Maintenance Needed (1)
No Maintenance (0)	1,892(TN)	8 (FP)
Maintenance Needed (1)	397 (FN)	7,703 (TP)

Table 3. Performance Comparison With and Without Reported Issues

Model Configuration	Accuracy	Balanced Accuracy
With Reported Issues	95.95%	97.34%
Without Reported Issues	77.06%	85.15%

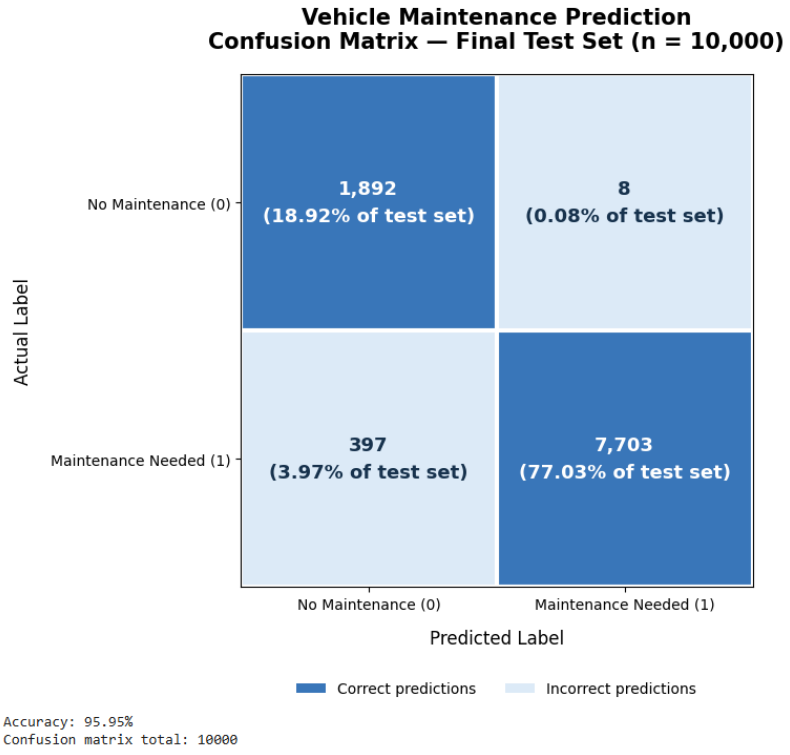


Figure 1. Final test-set confusion matrix

Classification Report (Final Test Set)

	precision	recall	f1-score	support
0	0.26	0.63	0.37	1900
1	0.87	0.58	0.70	8100
accuracy			0.59	10000
macro avg	0.56	0.60	0.53	10000
weighted avg	0.75	0.59	0.63	10000

Figure 2. Classification Report

5. Discussion

The model performed well on the test set after adding Reported_Issues and vehicle-condition features to the processing steps. A comparison made on the validation set showed that the model with Reported_Issues had 95.44% accuracy and 96.94% balanced accuracy. The model without this feature had 77.06% accuracy and 85.15% balanced accuracy.

This difference shows that Reported_Issues gives a lot of information in the data. However the link found needs to be looked at because all records with four or five reported issues were marked as needing maintenance in the data studied. This pattern could be a connection or it could be because of the way the target labels were created.

So the results should not be seen as proof that the model will do the same, in a real maintenance situation. The feature being available when making predictions, how the target labels were created and how the model does on a dataset should all be checked in future research.

6. Conclusion

In this project we created a vehicle maintenance classification model using Logistic Regression and a organized preprocessing pipeline. The final model achieved an accuracy of 95.95 percent and a balanced accuracy of 97.34 percent when tested on 10,000 records. The confusion matrix showed 9,595 predictions, which included 1,892 negatives and 7,703 true positives.

The results show that the model can classify the records in the selected dataset with performance. The link between Reported_Issues and the target variable needs research. Future work should focus on how the target labelsre generated test the model on new data compare other algorithms and consider the real-world costs of false positives and false negatives. These steps would help determine if the model is suitable, for world predictive maintenance.

7. Recommendations / Implications

1. Look into how the target called Need_Maintenance was created and write down any rules that are known for forming the labels.

2. Compare models that have Reported_Issues with models that do not have Reported_Issues and explain how their behavior differs.

3. Test the model using data that was collected in a way than the original dataset.

4. Focus on negatives because these are vehicles that actually needed maintenance but the model said they did not need it.

5. Use the method of splitting the data and evaluating it to compare Logistic Regression with types of classifiers.

6. Build a Streamlit tool to display predictions and ensure it is clear that this is a research example.

7. Keep track of how accurate the predictionsre check whether the model is good enough, before thinking about using it in real life.

8. Limitations of the Study

The study has limitations. First the method used to generate the target labels could not be confirmed because the dataset information did not provide details. Second the variable Reported_Issues shows a link with Need_Maintenance. In fact in the observed dataset values four and five in Reported_Issues perfectly match the corresponding values in Need_Maintenance. This perfect association raises concerns about target-construction bias or data leakage. However it does not prove that leakage actually happened.

Third the dataset has a target distribution. 81% Of the records belong to class 1 which could skew model performance. To address this both ordinary accuracy and balanced accuracy were reported. Fourth the model was tested on a held-out part of the dataset. This means the results do not show how well the model would perform on vehicles from populations, regions or operating conditions. Finally if the model were used in real-world settings it would be important to confirm that all input features are available before any maintenance decision is made. It would also be necessary to ensure that the predictions are reliable enough, for use.

9. References

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